

MURAT FURKAN BAYRAM

Founder & Lead Autonomous Systems Architect | TRUSTIA AI

Email: kariyer@trustia.com.tr | Phone: +90 537 064 04 60 | Location: Istanbul, Turkey | LinkedIn: linkedin.com/in/muratfurkanbayram | Web: trustia.com.tr

EXECUTIVE SUMMARY

Systems Architect and Deep Tech Entrepreneur with specialized expertise in GPS-denied autonomous vehicle guidance, 3D Pose Graph SLAM, multi-modal tactical sensor fusion, and certifiable deterministic robotics software. Designed and deployed TRUSTIA AI, a 100% zero-dependency, military-grade autonomous navigation brain with 1,276 automated validation tests (100% green pass) and full compliance with NATO STANAG 4586 and SAE JAUS interoperability standards.

OFFICIAL ACCREDITATIONS & CREDENTIALS

- Turkish Defense Industry Agency (SSB) & BTK Academy:** Defense Industry Score 100/100 (Credential ID: L2zPtN4X1ZJ).
- Ministry of Industry & Technology (KOSGEB):** Advanced Entrepreneurship Certificate (Ref: KSB01UGE0115153370).
- Scientific and Technological Research Council of Turkey (TUBITAK):** ARBIS Registered Researcher (ID: TBTK-0229-6571).
- ASELSAN Defense Systems:** Registered Supplier Identification Code (0050569CCE941FD1A49FCEFB9B7BE7D6).

PROFESSIONAL EXPERIENCE & TECHNICAL LEADERSHIP

TRUSTIA AI — Istanbul, Turkey & Abu Dhabi / International

Founder, CEO & Lead System Architect (2024 – Present)

- Autonomous Software Core Architecture:** Engineered a 15,401 LOC production-grade codebase with 100% zero external runtime dependencies, passing all 9/9 AST architectural safety and certification checks.
- GPS-Denied 3D SLAM & Odometry:** Implemented 3D Pose Graph SLAM with sub-10cm drift mitigation, visual-inertial odometry, and real-time point-cloud map fusion for electronic warfare contested environments.
- Tactical Threat Detection Fusion AI:** Developed multi-modal sensor fusion algorithms combining thermal imaging, metal induction, and Ground Penetrating Radar (GPR) to isolate IEDs and landmines into 30m dynamic hazard perimeters.
- Military Standards Compliance:** Implemented native protocol handlers for SAE JAUS (AS6009/AS6091), STANAG 4586, ROS 2, and CAN 2.0B / CAN FD industrial telemetry buses.
- Autonomous Swarm Coordination:** Designed decentralized consensus and formation control protocols for heterogeneous air-ground unmanned vehicle fleets.

TECHNICAL SKILLS & DOMAIN EXPERTISE

Robotics & Autonomy:	3D SLAM, Pose Graph Optimization, Visual Odometry, Path Planning (Hybrid A*, DWA), Swarm Intelligence
Sensor & AI Fusion:	LiDAR 3D, Radar GPR, Thermal Flir, Metal Induction, Sensor Fusion (Kalman/EKF), Multi-Modal AI
Standards & Protocols:	SAE JAUS (AS6009/6091), NATO STANAG 4586, ROS 2, CAN FD, Zero-Dependency Deterministic Systems
Languages & Tools:	Python (Pure stdlib & algorithms), C/C++, Embedded Linux, Git, Pytest (1,276 automated test suite)